

# Week 1 — What Do Engineers Actually Do All Day?

**Goal:** Replace the movie version of “engineer” with the real one, and figure out which flavor — mechanical or electrical — actually fits how you like to think and work.

## Part 1: Research (aim for 3–4 hours across the week)

Build a comparison table for **mechanical engineering (ME)** and **electrical engineering (EE)** with these rows. Every cell needs a source.

| Question                                                           | Mechanical | Electrical |
|--------------------------------------------------------------------|------------|------------|
| What problems does this field solve? (3 concrete examples each)    |            |            |
| What does a typical Tuesday look like for a working engineer?      |            |            |
| What industries hire the most of them near me?                     |            |            |
| Median pay in my metro area (entry level AND 10-year)              |            |            |
| What’s the most math-heavy part of the job?                        |            |            |
| What tools/software do they live in? (CAD? MATLAB? Oscilloscopes?) |            |            |
| What do people in this field complain about?                       |            |            |

### Where to look:

- U.S. Bureau of Labor Statistics *Occupational Outlook Handbook* (bls.gov/ooh) — search “mechanical engineers” and “electrical engineers.” Use the “Similar Occupations” links too.
- Reddit: r/MechanicalEngineering and r/ElectricalEngineering — search “day in the life” and “what I wish I knew.” Complaints are data.
- YouTube: search “day in the life mechanical engineer” (skip anyone selling a course).
- O\*NET Online (onetonline.org) — the “Work Activities” section is the unvarnished truth.

**Bonus research (optional, +respect):** Find one engineer on LinkedIn or YouTube whose actual job sounds interesting to you. Write down their title, company, and what caught your attention. This becomes useful in Week 7.

## Part 2: Essay (600–900 words)

**Prompt:** “Which of these lives do I actually want, and why?”

Requirements:

- Take a position. “Both seem cool” is not a position. It’s fine to be 60/40 — say so and say why.
- Use at least **4 specific facts** from your research table, cited.

- Include one honest paragraph about what worries you or seems hard about the field you're leaning toward. (Pretending there are no downsides is how people pick wrong.)
- End with one question you can't answer from the internet — something you'd have to ask a real engineer. Keep it; you'll use it in Week 7.

## Done when

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- ☐ Comparison table complete, every cell sourced
- ☐ Essay written
- ☐ 20-minute walkthrough conversation done